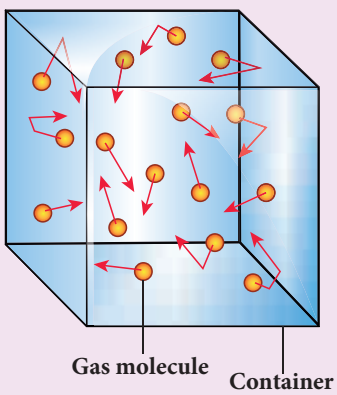
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6. These collisions are perfectly elastic so that there is no loss of kinetic energy during collisions.
7. Between two successive collisions, a molecule moves with uniform velocity.
8. The molecules do not exert any force of attraction or repulsion on each other except during collision. The molecules do not possess any potential energy and the energy is wholly kinetic.
9. The collisions are instantaneous. The time spent by a molecule in each collision is very small compared to the time elapsed between two consecutive collisions.
10. These molecules obey Newton's laws of motion even though they move randomly.

9.2 PRESSURE EXERTED BY A GAS

9.2.1 Expression for pressure exerted by a gas

Consider a monatomic gas of N molecules each having a mass m inside a cubical container of side *l* as shown in the Figure 9.1 (a).



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Figure 9.1 (a) Container of gas molecules

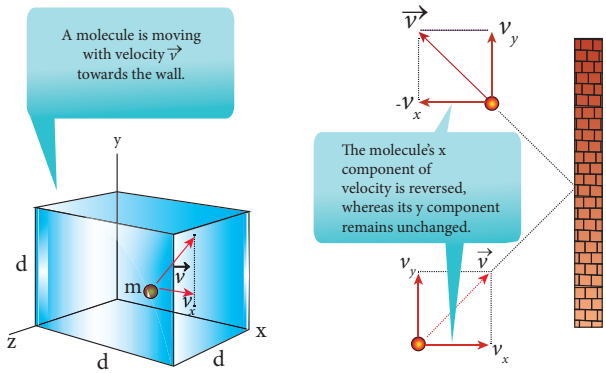


Figure 9.1 (b) Collision of a molecule with the wall

The molecules of the gas are in random motion. They collide with each other and also with the walls of the container. As the collisions are elastic in nature, there is no loss of energy, but a change in momentum occurs.

The molecules of the gas exert pressure on the walls of the container due to collision on it. During each collision, the molecules impart certain momentum to the wall. Due to transfer of momentum, the walls experience a continuous force. The force experienced per unit area of the walls of the container determines the pressure exerted by the gas. It is essential to determine the total momentum transferred by the molecules in a short interval of time.

A molecule of mass m moving with a velocity $\vec{v}$ having components (v_x, v_y, v_z) hits the right side wall. Since we have assumed that the collision is elastic, the particle rebounds with same speed and its x-component is reversed. This is shown in the Figure 9.1 (b). The components of velocity of the molecule after collision are ($-v_x$, v_y, v_z).

The x-component of momentum of the molecule before collision = mv_x